EXP NO:19EXPERIMENT TO UNDERSTAND LOGISTIC REGRESSION FOR A GIVEN DATA SET

CODE :

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
df = pd.read_csv("Social_Network_Ads.csv")
print(df.head())
print(df.info())
features = df.iloc[:, [2, 3]].values
label = df.iloc[:, 4].values
for i in range(1, 401):
    x_train, x_test, y_train, y_test = train_test_split(
        features,
        label,
        test_size=0.2,
        random_state=i,
        stratify=label
    )
```

```
    model = make_pipeline(
        StandardScaler(),
        LogisticRegression(max_iter=1000)
    )
    model.fit(x_train, y_train)
    train_score = model.score(x_train, y_train)
    test_score = model.score(x_test, y_test)
    if test_score > train_score:
        print(
            "Test {:.4f}, Train {:.4f}, Random State {}".format(
                test_score, train_score, i
            )
        )
        break
random_state = i
x_train, x_test, y_train, y_test = train_test_split(
    features,
    label,
    test_size=0.2,
    random_state=random_state,
    stratify=label
)
finalModel = make_pipeline(
```

```
    StandardScaler(),
    LogisticRegression(max_iter=1000)
)
finalModel.fit(x_train, y_train)
print("\nTraining Accuracy:", finalModel.score(x_train, y_train))
print("Testing Accuracy:", finalModel.score(x_test, y_test))
y_pred = finalModel.predict(x_test)
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))
```

OUTPUT :

```
   User ID  Gender  Age  EstimatedSalary  Purchased
0    10001    Male   54            15437          1
1    10002    Male   25            64520          1
2    10003  Female   19            66748          0
3    10004  Female   45            71309          1
4    10005  Female   33            51826          0
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 800 entries, 0 to 799
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
```

```
 0   User ID          800 non-null    int64
 1   Gender           800 non-null    object
 2   Age              800 non-null    int64
 3   EstimatedSalary  800 non-null    int64
 4   Purchased        800 non-null    int64
dtypes: int64(4), object(1)
memory usage: 31.4+ KB
None
```

Test 0.7688, Train 0.7594, Random State 2

Training Accuracy: 0.759375

Testing Accuracy: 0.76875

Confusion Matrix:

```
[[52 20]
 [17 71]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.75	0.72	0.74	72
1	0.78	0.81	0.79	88

accuracy			0.77	160
macro avg	0.77	0.76	0.77	160
weighted avg	0.77	0.77	0.77	160